



unilog | H2S

UniHack

Team Details

Team name: SKavengers

Team leader: Sumit Kumar Das

Challenge: AI-Powered Product Intelligence for Industrial Commerce

AI-Powered Product Intelligence for
Industrial Commerce

Brief about your solution

GlassBox

Six columns in, 252 out — and every one of them can show you its evidence.

A self-bootstrapping enrichment engine for industrial catalogues. It induces its own brand and attribute vocabularies from the corpus it is given, records the exact characters behind every value it emits, and refuses to invent the ones it cannot derive.

Runs with no API key, no network and no model download — so it cannot fail on the evaluation dataset.

100%

Invoice $\leq$ 40 chars

100%

Vocabulary compliance

9/9

Gold channels exact

214/s

Rows per second

1. How does it enrich minimal product information?

One real row of the supplied dataset, start to finish. Every value below is read from the generated delivery sheet.

INPUT - 6 COLUMNS

```
Mfg_Part_Num 49-94-1940
Part_Desc    49-94-1940 Milw 14"x1/8"x1" Masonr
              y Cut Off Disc
E1_Brand     -- Unbranded --
Unilog_Brand -- No Unilog Brand --
DIB_Brand    -- No DIB Brand --
Part_Manuf   Milwaukee Accessory (4031)
```

3 of the 6 columns are placeholders meaning "empty". 799 of the 1000 rows carry one; all 1000 carry another.

OUTPUT - 252 COLUMNS (extract)

```
BRAND_NAME
Milwaukee®

MANUFACTURER_NAME
Milwaukee Electric Tool Corporation

Classpath
Tools & Equipment > Power Tool Accessories > Cut-Off Wheels

Attributes
Diameter 14 in · Thickness 1/8 in · Arbor Size 1 in · Size 14 in x 1/8 in x 1 in

INVOICE_DESC
CUTOFF WHEEL 14IN 1/8IN 1IN MASONRY

MOBILE_DESC
Milwaukee, Cut Off Wheel, 49-94-1940, 14 in Diameter, 1/8 in Thickness
```

Every value is traced to its source characters. The run 14"x1/8"x1" is read positionally - diameter, thickness, arbor - because on a cut-off wheel that order is fixed, and swapping them would silently corrupt the record.

2. How does it ensure accuracy and trust?

Five mechanisms, all measured. Nothing below is estimated.

9/9

Gold channels
reproduced exactly

100%

Values inside the
controlled vocabulary

100%

Char-limit
compliance

0

Unit-format
violations

100%

252-header schema
conformance

Provenance, not self-report Every cell carries its mechanism, rule, confidence and source span. Confidence is computed from that distribution, so fluent invented values cannot score well.

Controlled generation Attribute values are constrained to a per-category vocabulary. A value outside it is rejected and reported, never emitted.

Verified against the published ground truth All five description channels of both fully-worked rows reproduce character-for-character, including a 390-character long description. Run `tests/test_gold_channels.py`.

Human-in-the-loop triage 618 rows auto-publishable, 297 flagged with a specific reason, 85 blocked. A reviewer clears an item in seconds.

It refuses to invent Part numbers, prices, UPC and country of origin are left blank with a recorded reason naming the system of record. Both gold rows leave the same fields blank.

3. What makes it scalable for enterprise catalogues?

214

rows/sec, single
threaded CPU

5s

for the full
1,000-row file

0

API keys or network
calls required

51

brands learned with
no dictionary given

Large catalogues Linear, single-pass, no per-row network call. 1,000 rows in 5 seconds on one core; the work parallelises per row, so a 2M-SKU catalogue is an overnight batch, not a budget line.

New manufacturers The vocabulary is induced from whatever corpus is loaded, so a new distributor needs no dictionary work. This is the whole point: a pipeline that depends on a hand-curated 27,000-row brand list can only process catalogues someone already curated.

Different formats The delivery schema is read from the organisers' own sheet at runtime, and input is accepted as CSV or XLSX in any column order. A changed header list is absorbed, not a code change.

Continuous updates Deterministic and idempotent: the same row always produces the same record, so re-runs are diffable and only genuinely changed rows need re-review.

Cost The core path uses no paid inference at all. The optional hosted layer is capped, cached, and only ever invoked for rows that need it.

Opportunities — how this differs, and its USP

The expected submission is: pandas → fuzzy match → one large LLM prompt → before/after view. Four things separate ours.

1 It learns its own vocabulary

The challenge is built on seven reference workbooks — a 27,000-row brand list, a 161,000-row list of values — and none of them are published on the portal. Teams following the guide literally will hardcode a few brands, and the brief warns that submissions must process an unseen evaluation set. We induce the lexicon from the corpus, so an uncurated catalogue works on day one.

2 Every cell is auditable

The challenge asks for traceable, explainable outputs. We take that literally: each of the 252 cells records the mechanism, the rule, the confidence and the exact source characters. A wrong value is traceable to the rule that produced it — the difference between a demo and something a content team can adopt.

3 It refuses to fabricate

Non-derivable fields are emitted blank with a reason naming the system of record. The Solution Guide states that a fluent description of invented values scores zero, and that reporting gaps is a strength.

4 It runs anywhere, deterministically

No key, no network, no model download; 240 rows/sec on a laptop. The optional AI layers attach on top and are never required, so the prototype cannot fail on the judges' machine.

List of features offered by the solution

ENRICHMENT

- Brand and manufacturer resolution via a 6-rung evidence ladder
- Dual taxonomy — Classpath plus Dept/Class/Fine, guaranteed consistent
- 17 per-category attribute contracts, 313 controlled values
- Trade-aware extraction: fractions, UOM, nominal sizes, wire codes

GENERATION

- Five description channels, each to its own house contract
- Two-sided constraint solver for the 60–80 char mobile window
- Hard 40-char all-caps invoice line with greedy-with-skip packing
- Deterministic digital-asset filenames and manufacturer URLs

TRUST

- Per-cell provenance with source character spans
- Confidence scored from provenance, not model self-report
- Triaged review queue with specific, actionable reasons
- Explicit refusal to populate non-derivable fields

OPERATIONS

- Zero-config: no key, no network, no model download
- Runtime schema binding — 252 headers read from the client sheet
- XLSX + CSV delivery, JSONL provenance sidecar, compliance report
- Vocabulary induction report and human-approval queue

Process flow

Two passes. The first learns the catalogue's vocabulary; the second enriches each row against it.

PASS 1 · LEARN

- Group by distributor account
- Score token distinctiveness (group TF-IDF)
- Link aliases to attested brands
- Induce series, product types, unit spellings
- Fit taxonomy + brand priors

Output: 51 brands, 181 held for review

PASS 2 · ENRICH EACH ROW

1. **Clean** — drop placeholders, expand shorthand
2. **Resolve entity** — 6-rung brand ladder
3. **Classify** — 87 leaves, both hierarchies
4. **Fill contract** — extract with source spans
5. **Normalise units** — fractions, approved UOM
6. **Render 5 channels** — constraint-solved lengths
7. **Score + triage** — confidence from provenance

DELIVERABLES

enriched.xlsx

252 columns, review rows tinted

provenance.jsonl

every cell's evidence

review_queue.csv

worst rows first, with reasons

report.md

compliance + evaluation

Optional layers attach at step 4 and are never required for a complete run.

Architecture

Deterministic core, optional AI on top. The core is complete on its own — which is why it cannot fail on an unseen dataset.

OPTIONAL · AI ACCELERATION

NVIDIA NIM proposals — validated against the vocabulary, labelled, always routed to review

Distilled local models — classifier + span tagger, CPU inference, zero marginal cost

CORE ENGINE · DETERMINISTIC, OFFLINE

induce · entity · taxonomy · attributes · extract · units · render · confidence

Vocabulary induction → evidence ladder → contract fill → constraint-solved generation

FOUNDATION

schema — 252 headers bound from the client sheet at runtime

provenance — every cell carries mechanism, rule, confidence, source span

Data flow: 6-column CSV/XLSX → learn → enrich → 252-column delivery sheet + provenance sidecar + review queue + compliance report

Technologies used in the solution

Core Python 3.12 · pandas · RapidFuzz · XlsxWriter · openpyxl

13 purpose-built modules, no ML framework in the critical path

NLP / IE Rule-based information extraction · group-conditional TF-IDF induction · fuzzy entity resolution · BIO span labelling

Constraint satisfaction for the length-bounded channels

Generative AI (optional) NVIDIA NIM — OpenAI-compatible endpoint, Llama 3.3 70B

Schema-constrained proposals, validated and cached; never on the critical path

ML (optional) PyTorch · Hugging Face Transformers · DistilBERT

Rule-engine distillation: 77-class classifier + 38-type span tagger

Interface Streamlit · custom CSS

Five views: enrich, compliance, review queue, vocabulary, export

Engineering pytest · git · deterministic seeds

Gold-row exact-match tests; every published metric reproducible from the CLI

The middle panel is the differentiator. The “Read from” column shows, for every extracted attribute, the exact substring of the raw description it came from — alongside the mechanism that produced it and its confidence. Blank cells state why they are blank.

Current status and next steps

SHIPPED AND VERIFIED

- End-to-end engine — 1,000 rows in 5s, 79% classified, 76% brand resolved
- 9/9 gold description channels exact · 100% vocabulary compliance · 100% schema conformance · 0 unit-format errors
- Provenance sidecar, triaged review queue, compliance report, Streamlit application

BUILT, PENDING A TRAINING RUN

The distillation package is complete and validated — 3,591 classifier examples over 77 classes and 636 span examples, generated from the rule engine, with the label-alignment and metric logic unit-tested (21 checks, including a pass over all 560 real span rows). The fine-tuning run itself is not yet executed, so no trained-model accuracy is claimed anywhere in this deck.

ROADMAP

- Manufacturer-source retrieval, honouring the guide's sourcing hierarchy, to close the attribute gap the input genuinely cannot supply
- Active learning — accepted review decisions feed straight back into the induced vocabulary
- Per-customer vocabulary packs, so an induced lexicon becomes a reusable asset per distributor

Links

GITHUB PUBLIC REPOSITORY

<https://github.com/skdas20/UniHack>

WORKING PROTOTYPE

<https://glassbox-unihack.streamlit.app/>

DEMO VIDEO (3 MINUTES)

Submitted alongside this deck on the Hack2skill portal

Run it yourself: `git clone · pip install -r requirements.txt · python run.py` — no API key, no network, ~5 seconds for 1,000 rows.

unilog | **H2S**
HACK2SKILL

UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Thank You

